CMR ENGINEERING COLLEGE

UGC AUTONOMOUS

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III-B.TECH II-SEM (R22) I-MID Examinations-February-2026 **Date: 16/02/2026**

Subject: Machine Learning **Time: 01:30 to 03:30 PM**

Branch: CSE, IT, CSD& CSM **Marks: 30 M**

Answer all Questions in Par-A & Answer any FOUR Questions in Part-B

PART-A **5x2M = 10M**

		BTL	CO
1.	Define the concept learning and version space?	**1**	**1**
2.	What are the issues in Machine Learning?	**2**	**1**
3.	Given a perceptron with inputs x1=1,x2=0, weights w1=0.5,w2=-0.4, and threshold θ=0.2,compute the net input and determine the output(use step	**3**	**1**
4.	Explain the sigmoid output for the net input=0.	**3**	**2**
5.	What is Entropy and information gain	**1**	**3**

PART-B **4x5M= 20M**

		BTL	CO
6.	Explain about Designing a Learning system	**6**	**1**
7.	Explain concept learning and describe the Candidate elimination algorithm using the following **Enjoy Sport** training dataset.	**2**	**2**

Example	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
1	Sunny	Warm	Normal	Strong	Warm	Same	**YES**
2	Sunny	Warm	High	Strong	Warm	Same	**YES**
3	Rainy	Cold	High	Strong	Warm	Change	**NO**
4	Sunny	Warm	High	Strong	Warm	Change	**YES**

8.	Briefly explain about the Back propagation algorithm and derivation?	**1**	**2**
9.	Write a short notes on the following a. Radial Basis functions. b. The Brain and Neuron	**3**	**2**
10.	Describe about Support Vector Machine	**2**	**2**
11.	Explain about Decision tree learning with example	**3**	**3**